D.4.2. Project Documentation: Design

Project: Canadian Labour Market Analytics: Industry Growth and Provincial Employment Trends, 2019–2025

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1. Introduction and Context

1.1 Background and purpose

The Canadian Labour Market Analytics project will examine employment growth, decline, and recovery across provinces and industries. Its central research question is: Which industries are growing in which Canadian provinces, and where has employment strengthened or recovered over time? The report will support students, recent graduates, job seekers, career advisors, employers, recruiters, educational institutions, researchers, business analysts, and economic-development organizations.

This design document explains how the project will acquire, prepare, integrate, store, and analyse Canadian employment data. It translates the project requirements into system architecture, a relational data model, component logic, and report interfaces. It also defines the technical constraints, security controls, and design decisions that will guide implementation and testing.

1.2 Scope and source context

The numerical analysis will cover January 2019 to December 2025, an 84-month period, across the ten Canadian provinces. The full CSV for Statistics Canada Table 14-10-0355-01 will be filtered to Statistics = Estimate and Data type = Seasonally adjusted (Statistics Canada, 2026). Employment will remain in thousands of persons, with one fact record per month, province, and selected industry.

Source	Role in the project	Integration approach
Statistics Canada employment CSV	Provides monthly employment estimates, reference periods, source identifiers, units, and status flags.	Supplies employment facts and identifiers used to resolve province, industry, and month keys.
WDS JSON metadata — product 14100355	Provides dimension definitions, member identifiers, labels, and available classification information.	Maps source geography and industry identifiers to their corresponding dimension records.
Scraped *The Daily* releases	Provides article titles, publication dates, reference	Connects articles to reference months and manually reviewed province and

	months, body text, and source links.	industry tags through separate article and bridge tables.

1.3 Requirements addressed by the design

The design builds on the project's User Requirements and Analysis Document. The original functional requirement identifiers (FR) are retained below to connect each requirement to its proposed implementation.

Requirements	Design coverage
FR-01–FR-05	Acquisition, metadata mapping, scraping, preparation, and integration
FR-06	MySQL schema, keys, loading, and validated reporting tables
FR-07–FR-11	Filters, employment measures, changes, rolling average, and recovery
FR-12, FR-14, FR-15	Matrix, trend chart, industry drill-through, and bookmark toggle
FR-13	Article relevance, text search, excerpts, and source links

1.4 Analytical rules and interpretation

The following rules define how the system will calculate and present employment results. They also provide the basis for independent calculation checks during testing.

Employment levels and aggregation

Employment will be reported in thousands of persons for a selected reference month. Monthly estimates will not be summed to create an employment level for a date range. Industry comparisons will use compatible, non-overlapping categories; parent totals will not be added to their component industries. Missing or suppressed estimates will remain null, with source flags retained, and will be shown as unavailable rather than zero.

Monthly and annual changes

Month-over-month change compares the selected month with the immediately preceding calendar month. Year-over-year change compares it with the same month one year earlier. Absolute change equals current employment minus comparison employment. Percentage change equals the absolute change divided by comparison employment, multiplied by 100. A missing required estimate makes the result unavailable; a zero comparison estimate also makes percentage change unavailable. These outcomes will be distinguished from a valid zero change.

Twelve-month rolling average

The rolling average will use the selected month and the preceding 11 calendar months. All 12 estimates must be valid. The calculation will not skip missing months or substitute zero. A required month outside the acquired study window will make the result unavailable.

Employment recovery

The recovery index equals the selected month's employment divided by February 2020 employment, multiplied by 100. A missing or zero baseline makes the index unavailable. Sustained

recovery is a project-defined measure: after an observed post-February 2020 decline below the baseline, it is the first month of the earliest sequence of three consecutive valid months at or above the baseline. A missing month interrupts the sequence.

The report will distinguish sustained recovery, no observed decline, no recovery observed, and insufficient data. A data gap that prevents classification will not be interpreted as evidence of no decline or no recovery. This operational definition is specific to the project and is not an official Statistics Canada classification.

Sustained recovery will be evaluated using observations from March 2020 through the selected reference month. All three qualifying months must fall within that interval. The classification will be unavailable for reference months before March 2020.

Article context

Articles will be linked to their reference months and reviewed province and industry tags. Publication dates will be kept separately. Article text will support interpretation without replacing the numerical CSV estimates; identified differences will be documented. Multiple article associations must not duplicate employment values, and the report will indicate when no supporting article is available.

2. Goals and Non-goals

2.1 Goals and measurable success criteria

These targets will be tested after implementation. All functional requirements remain in scope. Identifiers link to the original functional requirements (FR), non-functional requirements (NFR), and acceptance criteria (AC).

Design goal	Measurable success criterion	Requirement ID
Correct scope and mixed sources	Retain only the selected estimate series and ten provinces within the 84-month calendar; document missing coverage. Parse the JSON metadata and successfully scrape at least one article from a recorded target list.	FR-01–03; AC-01–03
Accurate and consistent data	Match 100% of at least 30 reproducibly sampled fact rows to source records. Find zero duplicate fact-grain keys and zero orphan foreign keys. Article links must not inflate employment totals.	NFR-01–02; AC-05
Reliable analytical measures	Match independent calculations for employment, absolute and percentage MoM/YoY changes, the complete 12-month average, and recovery against February 2020. Verify missing values, zero denominators, and all four recovery classifications.	AC-08–11

Design goal	Measurable success criterion	Requirement ID
Responsive report	Across ten repeat runs, each core page loads within five seconds and standard filter interactions complete within three seconds in at least nine runs. Record hardware, software versions, and dataset size.	NFR-03
Usable exploration	Three reviewers complete filtering, a YoY comparison, and industry drill-through unaided, and identify the displayed unit and comparison period correctly.	NFR-06
Complete interactive functionality	Verify the matrix and trend values against their measures; confirm that drill-through carries the selected industry, bookmarks preserve filter selections, and article searches return relevant excerpts and working source links.	FR-07; FR-12–15; AC-07; AC-12–15
Controlled and traceable processing	A read-only account cannot INSERT, UPDATE, DELETE, or ALTER prepared tables. Preserve source identifiers, retrieval details, raw inputs, and transformation records; attribute sources and log retrieval and data-quality failures.	NFR-04–05; NFR-08–09
Reproducible delivery	A team reviewer rebuilds the prepared tables from the runbook and refreshes Power BI successfully. Validation totals, data types, scale, and selected SQL comparisons match the report.	NFR-07; NFR-10

2.2 Non-goals

The design excludes the three territories, city-level analysis, employment forecasting, individual job advertisements, salaries, applicant records, causal modelling, and real-time or automatically scheduled updates.

Employment growth will not be treated as a direct measure of vacancies or individual hiring prospects. A production web application and public cloud deployment are outside the baseline course prototype.

Article coverage may be selective; months without supporting commentary will be labelled accordingly.

3. System Architecture

The system will use a batch workflow with six responsibilities: acquisition, preservation of source evidence, preparation and integration, database validation and storage, analytical calculation, and interactive reporting. Figure 1 shows the flow between these responsibilities; the project team initiates each acquisition and refresh.

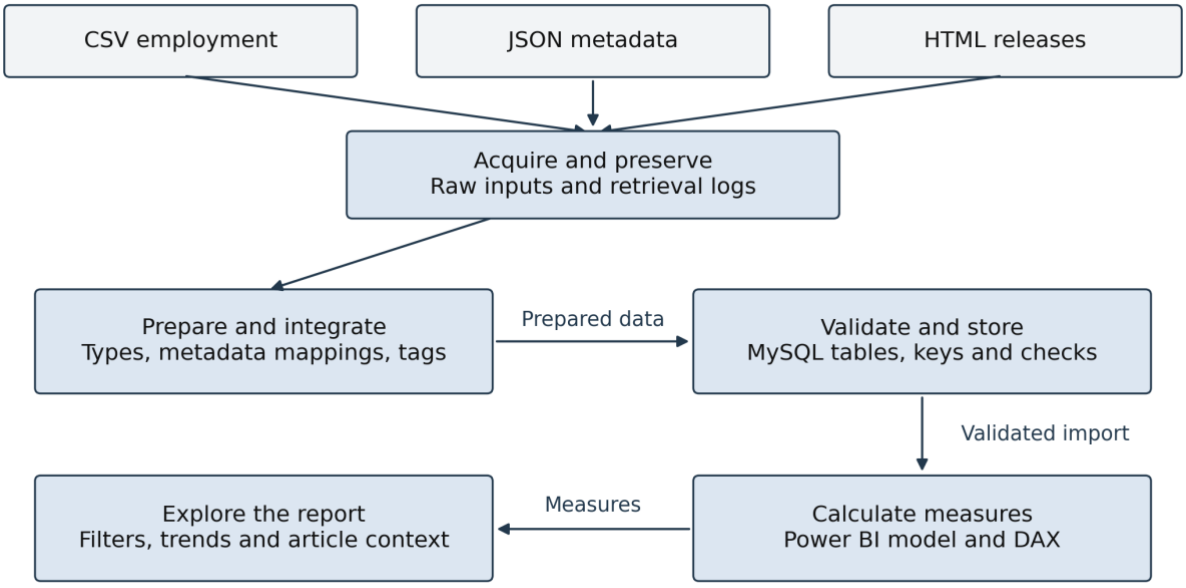


Figure 1. High-level system architecture of the Canadian Labour Market Analytics project.

3.1 Data flow and operation

The preparation workbook and database loader provide an explicit transfer between Power Query and MySQL. The final schema will be created using version-controlled SQL before data are loaded. This design uses the following sequence:

1. Acquire and preserve inputs. Download the complete table CSV and JSON metadata, scrape the documented list of article URLs, and save the raw files. Validate response formats so an HTML error page cannot be accepted as a CSV or JSON source. Keep failed retrievals in the acquisition log.

2. Prepare tabular outputs. Use Power Query in Excel to filter the study scope, transform employment data, expand metadata, and prepare article metadata and reviewed tag tables. Apply the scope filters before loading to worksheets and check output row counts. Query results can be loaded to worksheets (Microsoft, n.d.-a).

3. Export and load MySQL. Save each prepared worksheet as a separate UTF-8 CSV file, preserving the workbook and its query steps (Microsoft, n.d.-b). Empty employment values in exported CSV files will be loaded as SQL NULL, with the original source flags preserved. Full article bodies will be read directly from the saved UTF-8 text or JSON files. A Python loader will read both paths, resolve keys, and load the defined dimension, article, bridge, and fact tables.

4. Validate the database snapshot. Check scope, source-value samples, duplicate keys, missing values, referential integrity, and totals before publishing the new snapshot. Use a transaction for the final data changes so a failed load can be rolled back. Refresh the report only after the database checks pass.

5. Refresh the analytical model. Import the validated MySQL tables into Power BI, confirm relationships and data types, and apply DAX measures. Industry aggregation will use non-overlapping categories. Employment levels will refer to a selected month rather than a sum across months. Article filtering will not change the underlying employment totals.

6. Review and retain evidence. Test visual interactions, article search, independent calculations, and the performance and usability targets. Record the input snapshot, query and script versions, validation results, unresolved issues, and review decisions so another team member can reproduce the run.

The semantic model will retain the complete study window so comparison months and the February 2020 baseline remain available when a user narrows the displayed date range. A calculation requiring a month outside the acquired window, or a missing or suppressed value, will return unavailable under the rules in Section 1.4. Historical article figures will provide context and will not overwrite the selected numerical data snapshot.

4. Logical Architecture

The proposed batch system separates acquisition, preparation, validated storage, semantic modelling, and user reporting. Raw inputs and logs remain available for traceability (FR-01–06). The Power BI model never joins multiple article tags directly to employment facts, preventing inflated totals.

Input layer	Processing and storage	Presentation
Statistics Canada CSV; WDS JSON; The Daily HTML	Python retrieval and extraction → raw snapshots/logs	Power BI Import model
Industry and geography identifiers	Excel Power Query cleaning → UTF-8 exports → Python loader	DAX changes, rolling average, recovery
Article text and source links	MySQL facts, dimensions, articles, tag bridges	Matrix, trends, drill-through, article explorer

5. Physical Architecture

Baseline deployment: one Windows workstation runs Python, Excel Power Query, a local MySQL Server, and Power BI Desktop. External CSV and HTML downloads use HTTPS GET; Statistics Canada Web Data Service metadata uses HTTPS POST (getCubeMetadata).

Microsoft lists Windows among the Desktop platform requirements (Microsoft, 2026a). Access to this workstation and the required Excel installation must be confirmed before implementation. Team members using macOS may contribute scripts, source files, and documentation, with report authoring and final refresh performed in the agreed Windows environment.

The Python loader connects to localhost:3306 using a write-limited account; Power BI imports through a read-only MySQL account. GitHub holds code and instructions, not passwords or raw credentials.

Power BI will use Import mode through the MySQL connector, with Oracle MySQL Connector/NET installed as required by Microsoft (2026b).

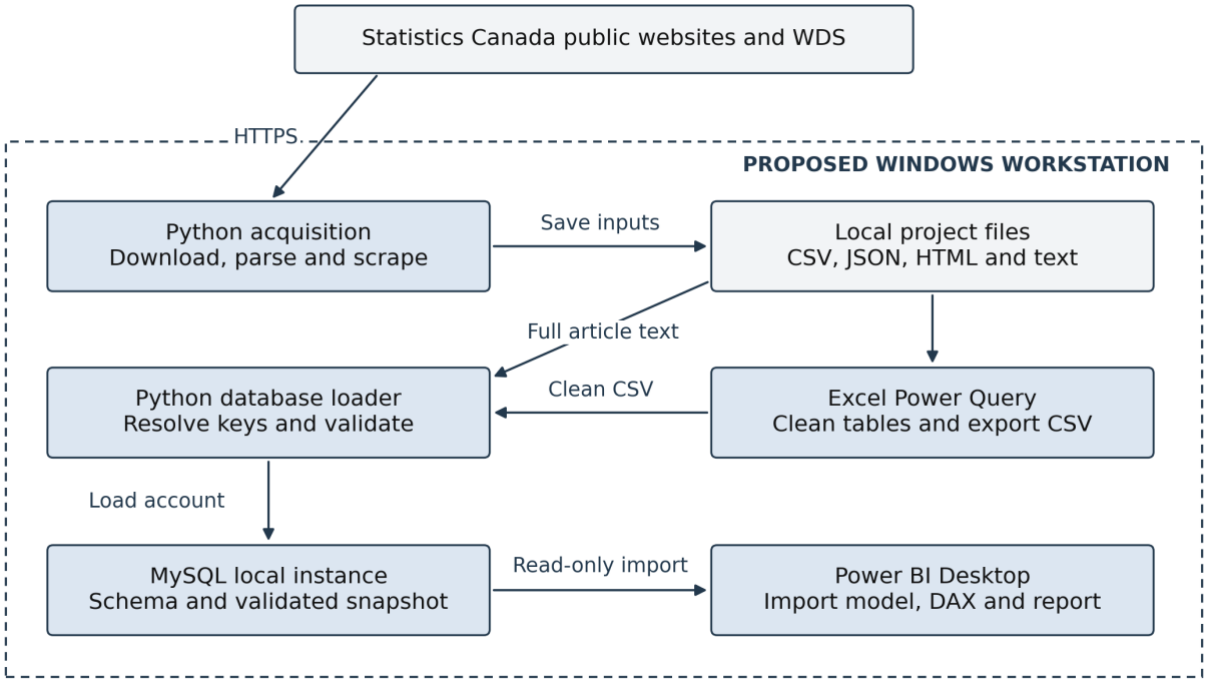


Figure 2. Proposed deployment and file transfer paths on the Windows workstation.

Location	Runtime / protocol	Output
Statistics Canada websites / WDS	HTTPS GET / POST	CSV, JSON, HTML
Windows project folder	Python; Excel Power Query; UTF-8 files	Clean tables and audit logs
Local MySQL + Power BI Desktop	MySQL connector; Import mode	Validated tables; interactive .pbix

The team will record workstation specifications and installed software versions before end-to-end integration and performance testing.

6. Data Model

6.1 ERD and relational schema

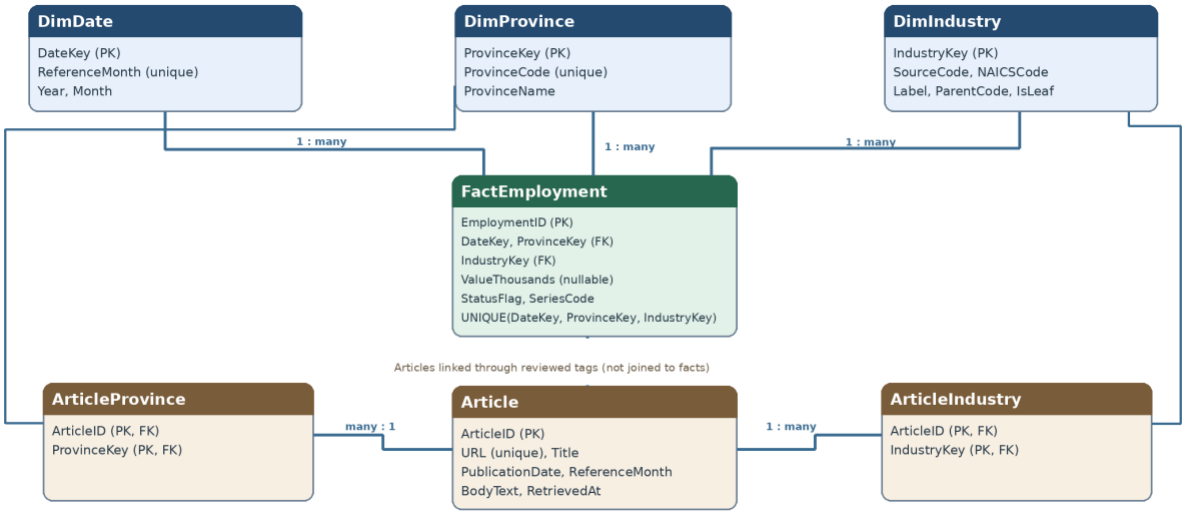


Figure 3. Proposed MySQL entity-relationship diagram (PK = primary key; FK = foreign key).

Each employment fact record will represent one reference month, one province, and one selected industry for the Estimate and Seasonally adjusted series. Dimension keys will link the fact table to the month, province, and industry dimensions. A unique constraint on DateKey, ProvinceKey, and IndustryKey will enforce this grain. The approved industry selection and analytical measures will prevent aggregation of overlapping parent and child categories.

Entity and primary key	Core attributes / foreign keys
DimDate [DateKey]	ReferenceMonth (unique), Year, Month
DimProvince [ProvinceKey]	ProvinceCode (unique), ProvinceName
DimIndustry [IndustryKey]	SourceCode, NAICSCode, Label, ParentCode, IsLeaf
FactEmployment [EmploymentID]	DateKey FK, ProvinceKey FK, IndustryKey FK, ValueThousands nullable, StatusFlag, SeriesCode; UNIQUE(DateKey, ProvinceKey, IndustryKey)
Article [ArticleID]	URL unique, Title, PublicationDate, ReferenceMonth, BodyText, RetrievedAt
ArticleProvince	Composite PK (ArticleID, ProvinceKey); both columns are foreign keys.
ArticleIndustry	Composite PK (ArticleID, IndustryKey); both columns are foreign keys.

ERD cardinality:

DimDate 1—∞ FactEmployment;

DimProvince 1—∞ FactEmployment;

DimIndustry 1—∞ FactEmployment;

Article 1—∞ ArticleProvince and ArticleIndustry;

each bridge ∞—1 corresponding dimension.

6.2 Data flow chart

Step	Input → process → output / control
1 Acquire	Official CSV / WDS / article URLs → Python validation and download → archived raw files + timestamp/error log
2 Prepare	Raw CSV / JSON → Power Query type, scope, NAICS and null handling → UTF-8 prepared tables; unmatched-code log
3 Load	Prepared CSV + full article text → transactional Python/MySQL load → dimension, fact, article, bridge tables
4 Validate	SQL tests: unique fact keys, FK orphans, source samples, totals, invalid zeros → accepted snapshot or rollback
5 Report	Read-only Power BI import → star relationships and DAX → filtered visuals / source-linked commentary

7. Component Design

7.1 Acquisition and preparation algorithm

Pseudocode: FOR source IN {CSV, WDS JSON, approved article URLs}: request via documented endpoint; verify HTTP status and content type; save original + URL + timestamp; IF unsuccessful, append error and continue. Filter CSV to ten provinces, study months and selected adjusted estimate; cast dates and values; retain suppressed values as NULL; split NAICS label when present; map identifiers using JSON; log duplicates and unmapped categories.

7.2 Loading and validation algorithm

BEGIN TRANSACTION; load Date, Province, Industry and Article dimensions; resolve surrogate keys; load one FactEmployment record per unique grain; populate reviewed article bridge records; reject or log unmatched keys; test duplicates, orphans, units, sample values, and row counts; IF every mandatory check pass COMMIT, ELSE ROLLBACK and log the cause.

7.3 Analytics and recovery logic

Employment = selected-month estimate in thousands, not the sum of estimates over months. MoM and YoY compare the prior month or same month of the prior year; % change = (current − prior) / prior × 100, blank if prior is zero/missing. Rolling 12M = arithmetic mean of exactly 12 consecutive valid months, otherwise blank. Recovery index = current / February 2020 × 100. Sustained recovery begins at the first month of the earliest 3-month valid sequence at/above baseline after an observed decline; gaps interrupt qualification. Distinguish no decline, no recovery and insufficient data (FR-08–11).

7.4 Test and failure behaviour

Tests cover source-to-target comparison (≥30 sampled rows), zero duplicate fact keys, zero orphan FKs, missing-value preservation, known calculation examples, source links and bookmark/filter state. Failed HTTP requests, absent months or unapproved tags are logged; unavailable estimates are never set to zero.

8. Interface Design

8.1 Power BI report wireframes

Page / compact wireframe	Interactive elements
OVERVIEW: [Province ▼] [Industry ▼] [Month ▼] [Employment] [MoM] [YoY] [Recovery] [Monthly employment / 12M line graph]	Slicers filter relevant visuals; show unit (thousands of persons) and selected reference month.
COMPARISON: [Absolute | YoY %] [Industry × Province conditional-format matrix]	Bookmark toggle swaps displayed measure and title/units; avoid overlapping industry categories.
INDUSTRY DETAIL: [Industry name] [Trend + rolling average] [Recovery] [Province]	Right-click drill-through preserves chosen industry and shows date and province comparisons.
ARTICLE CONTEXT: [Text search] [Month] [Article title | excerpt | source URL]	Search reviewed tagged articles; clear no-article state; open original publication links.

8.2 Interfaces, protocols, constraints and security

External interfaces: HTTPS GET for CSV/HTML and HTTPS POST for WDS JSON; local UTF-8 CSV/text interchange; SQL over MySQL localhost; Power BI MySQL Import connector. No public API or cloud hosting is required for the baseline prototype. Limits include workstation RAM/CPU, download size, incomplete historical series, source classification changes, scraper layout changes, and Power BI refresh latency. Store credentials outside GitHub, use least-privilege loader/report accounts, preserve public-source attribution and raw audit files, and avoid collecting personal job-seeker information.

9. Design Rationale

Decision	Justification and trade-off
Python + official sources	Automates repeatable metadata parsing, scraping, error logging; source structure may change.
Excel Power Query	Visible, repeatable cleaning and NAICS transformations; CSV export is an explicit handoff.
MySQL relational design	Enforces keys, snapshot integrity and article bridge separation; requires local setup.
Power BI Import + DAX	Efficient interactive matrix, drill-through and time calculations; refresh is manual.

Historical scope / data risk	Keep 2019–2025 target; Table 14-10-0355-01 is labelled “last 5 months,” so full coverage must be verified or source change approved before implementation.

Design status: planned architecture, not evidence of a deployed system. References and requirements traceability are retained in Sections 1–3 and the existing reference list.

References

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